

The Trip Economy: Which Foul-Drawing Skills Travel?

Introduction

A trip to the free-throw line is the most valuable scoring attempt in basketball. At league conversion, a two-shot trip yields 1.57 expected points, more than a shot from any zone on the floor. Yet foul drawing is measured coarsely. Free-throw rate treats all trips alike, and possession arithmetic approximates trip counts with the 0.44 coefficient. Recent work folded the free throws earned on fouled shots into shot value; the rest of the trip economy is countable with existing tools but unstudied as a skill. We ask how players actually get to the line and which channels are repeatable player skills rather than circumstance.

Methods

We reconstruct every regular-season NBA free-throw trip, meaning the free throws awarded from a single non-technical foul, from official play-by-play across the three most recent seasons (2023-24 through 2025-26, 3,690 games). We trace each trip to the foul that caused it and classify the trips into eight classes in two tiers. Attempt-equivalent trips replace a field-goal attempt as the possession's scoring attempt (two- and three-shot shooting fouls, bonus). Add-on trips stack points on a possession that already stands (and-ones and rarer classes). Reconstruction is exact rather than estimated. Every player-game reconciles with the official box score and every one of 1,723 league player-seasons with an independent season-totals source, yielding 88,347 trips with zero discrepancies. The dataset and pipeline are public.

Results

The trip economy varies structurally across the 284 players with 300 or more field-goal attempts in 2025-26 (Table 1); James Harden leads in three-shot fouls drawn (3.8 per 100). Channels persist differently year over year (422 player-season transitions; the pattern holds in both season pairs): two-shot shooting fouls $r = .87$, three-shot fouls $.76$, and-ones $.75$, bonus $.58$, residual classes $.31$. Two-shot fouls persist near their split-half reliability ceiling ($.89$ to $.92$); bonus trips fall short of a lower one ($.65$ to $.78$), noisier to measure and less stable (Figure 1). Changing teams lowers persistence in every channel: for 54 players who moved between seasons without a midseason change, all-trip persistence is $.74$ against $.87$ for 284 stayers ($p = .014$), with the largest drop in bonus ($.62$ to $.40$, $p = .046$). With 54 movers the bonus gap's bootstrap interval spans zero, so channel-specificity is suggestive, not established. The conventional 0.44 coefficient mismeasures the same players every year: its true value spans $.277$ to $.481$, persists at $r = .53$, and moves True Shooting by a median 0.15 and up to 1.5 percentage points (Table 1).

Conclusion

Drawing shooting fouls is the most repeatable route to the line; the non-shooting bonus channel is the least stable and the least portable across a team change. The split is actionable with care: shooting-foul generation projects across team changes, bonus volume should be projected with destination-team context and wider uncertainty, and exact trip accounting removes a persistent, player-shaped error from efficiency metrics.

Foul-drawing channels persist differentially, and the non-shooting bonus channel persists least
Each dot: the correlation between a player's rate in one season and the next, computed separately over player-season pairs where he stayed with one team (blue, $n = 284$) and where he changed teams between seasons (orange, $n = 54$); 84 pairs with a midseason move are excluded.

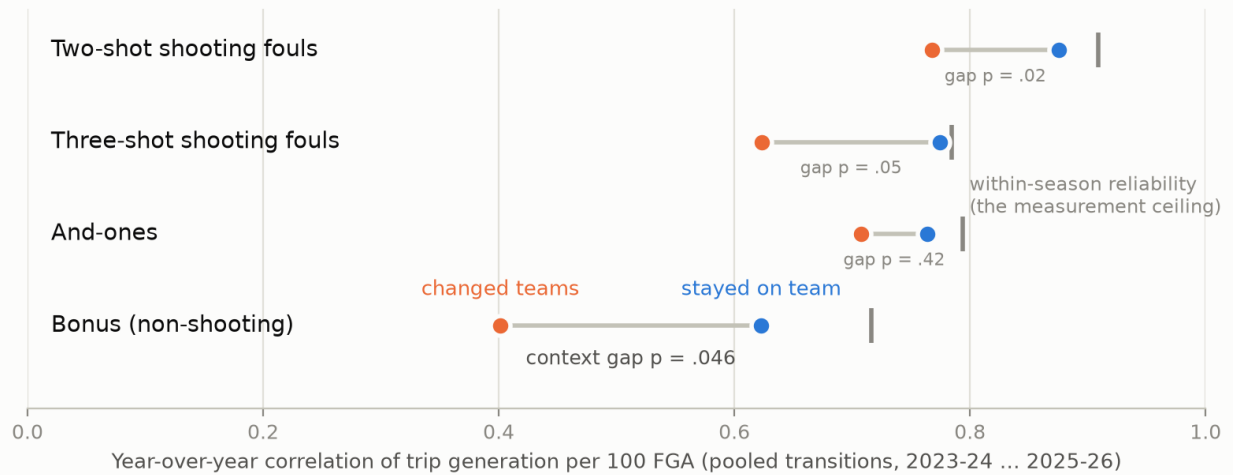


Figure 1. Year-over-year persistence by channel for players who stayed with one team vs changed teams, against each channel's within-season reliability; pooled over 2023-24 ... 2025-26.

Table 1. The trip taxonomy, measured. League trips count every 2025-26 trip; the rate distribution and split-half reliability are over the 284 players with ≥ 300 field-goal attempts in 2025-26; year-over-year r is pooled over 2023-24 ... 2025-26.

Channel	Tier	League trips (2025-26)	Median /100 FGA (p10–p90)	Year-over-year r	Split-half reliability
Two-shot shooting fouls	attempt-equivalent	18,846	8.1 (3.1–14.0)	0.87	0.91
Three-shot shooting fouls	attempt-equivalent	933	0.2 (0.0–1.1)	0.76	0.78
And-ones	add-on	5,860	2.6 (0.9–4.5)	0.75	0.79
Bonus (non-shooting)	attempt-equivalent	4,760	1.8 (0.8–3.7)	0.58	0.72
Other add-on (flagrant, away-from-play, transition, clear path)	add-on	549	—	0.31	—
All trips	—	30,948	—	0.86	0.95

Note. The true attempt-equivalent coefficient — the exact replacement for the conventional 0.44 — has median 0.427 across qualified players (p10–p90 0.385–0.455, full span 0.277–0.481) and persists year over year at $r = 0.53$.